

Yang Yu

Software Engineer / Backend, AI Applications & Distributed Systems
Davis, CA | devilsrocbuddhasgildedimage@gmail.com | +1-530-304-3656
linkedin.com/in/yy030305 | github.com/eyesofish

Summary

Software engineer and UC Davis M.S. in Computer Science candidate with experience in Python, Java, C++, SQL, JavaScript, REST APIs, AI/RAG workflows, distributed systems, and engineering reliability. Builds evidence-driven software projects and debugs large codebases, build pipelines, and test failures.

Skills

- **Languages:** Python, Java, C++, SQL, JavaScript
- **Backend & APIs:** FastAPI, Spring Boot, REST APIs, async API design, MySQL, Redis, Docker, Kubernetes
- **AI & Retrieval:** LangChain, LangGraph, agentic workflows, RAG, embeddings, vector retrieval, reranking, PyTorch, Qwen API
- **Systems & Engineering:** Protobuf/RPC, Raft consensus, distributed systems, Git, unit testing, CI/test triage, debugging, monorepo build/test workflows

Experience

- **Microsoft** Suzhou, Jiangsu, China
Software Engineering Intern Jun 2026 – Present
 - Integrated the Microsoft Copilot mobile codebase, using sparse checkout and prebuilt dependencies to establish local builds and unit tests for a large monorepo; validated authentication and a demo in an iPhone simulator.
 - Traced a macOS Keychain authorization loop, rebuilt the login keychain, cleared conflicting authentication processes, and installed team-pinned Apple IDE components to restore source access and resolve build-version mismatches.
 - Triageed about 30 reliability-pipeline builds, identifying six test targets as deterministic failures rather than flaky tests; supported isolation instead of deletion to reduce automation noise while preserving technical-debt tracking.

Projects

- **LLM + RAG Software Engineering Resource Recommender** Jan 2026 – Present
LangChain, LangGraph, deepagents, Qwen API
 - Built an agentic RAG workflow that decomposes technical queries into retrieval, filtering, reasoning, and result-organization stages, returning source-grounded recommendations for datasets, benchmarks, projects, and tools.
- **ZhihuRec 1M Recommendation System Prototype** Apr 2026 – May 2026
FastAPI, MySQL, Python, JavaScript, Docker
 - Built feed, search, event, profile, and debug REST APIs with multi-source retrieval, topic reranking, hot/fresh fallback, cold-start profile blending, and click-driven profile updates.
 - Validated search-to-feed feedback across 121 replay events, raising Post-Search Topic Carryover@10 from 0.9000 to 1.0000 (Gain@10 = +0.1000).
- **Apache Incubator ResilientDB – Raft Implementation** Sep 2025 – Nov 2025
C++, Bazel, Protobuf/RPC, Distributed Systems
 - Implemented a Raft path in a personal ResilientDB branch, covering leader election, heartbeat, log replication, commit flow, state-machine application, persistence, snapshots, and crash recovery.

Education

- **University of California, Davis** Davis, CA
M.S. in Computer Science (Expected May 2027) Sep 2025 – May 2027
- **Dalian University of Technology** Dalian, China
B.Eng. in Software Engineering Aug 2021 – Aug 2025